

UVD701: CAD for VLSI

L T P Cr

2 0 2 3.0

Course Objective: In this course the students will learn VLSI CAD tools and its related concepts and algorithms with emphasis on design automation. Students will also learning the scripting language used for design automation.

Introduction to VLSI Design: VLSI design flow, VLSI design styles, Design rules, Automation, use of VLSI CAD tools, Algorithmic Graph Theory, Computational Complexity and ROBDD.

Partitioning and Floor planning: Group-migration algorithm: KL algorithm, FM algorithm, Simulated Annealing and Evolution.

Pin Assignment, Placement, Routing and Compaction: Placement, Layout styles, Discrete methods in global placement, Timing-driven placement, Routing: Global Routing, detailed routing, Graph models, Line Search, Maze Routing, Channel Routing, Steiner Tree based Algorithms, ILP base approaches. One Dimension compaction, Two Dimension compaction, Hierarchical Compaction, Compaction Algorithms.

Performance Issues in circuit layout: delay models, timing driven placements, timing driven routing, Via Minimization, Over the Cell Routing – Single and Two layers routing, Clock and Power Routing.

Clock Synthesis and Synchronization: Synchronizers, Arbiters, Clock Synthesis, PLLs.

Perl: Introduction to Perl, Literals, Variable types and contexts, Perl Arrays, Perl Hashes, Operators, Conditionals, Loops & Subroutines, Files and File handling, Directory, Pattern Matching and Regular Expressions, Accessing System Resources, Generating Reports with Perl.

Laboratory Work:

Modelling various algorithms for partitioning, floor planning, placement, and routing. Performance metrics extraction of circuit layout. Writing basic program in Perl.

Course Learning Objectives (CLO)

The students will be able to:

1. Acquire knowledge about CAD tools used for VLSI design.
2. Able to understanding Algorithms for VLSI Design Automation.
3. Understand Timing Analysis.
4. Able to write script for design automation.

Text Books

1. Sherwani, N., Algorithms for VLSI Physical Design Automation, Springer (2005) 3rd ed.
2. Trimberger, S. M., An Introduction to CAD for VLSI, Kluwer (1987).
3. Sait, S. M. and Youssef, Habib, VLSI Physical Design Automation – Theory and Practice, World Scientific, 2004.
4. Ellie Quigley, Perl by Example, Prentice Hall, 5th edition, 2014

Reference Books

1. Gerez S.H., Algorithms for VLSI Design Automation, John Wiley (1998)
2. Sarrafzadeh, M. and Wong, C. K., An Introduction to VLSI Physical Design, McGraw Hill (1996).